

Post-Detection report for 2022 IAA SETI Permanent Committee meeting, 20 Sept 2022, Paris / online

Report given remotely by Kathryn Denning. (Associate Professor, Anthropology, York University, IAA SETI Committee member since 2005, SETI Institute Science Advisory Board member)

Prefatory note

We have not had a formally organized Working Group lately, but nonetheless some members (and researchers outside the committee) have been busily working on post-detection-related topics over the past year. Below I provide a brief summary of some of those developments since the last update at the 2021 IAC. (This summary isn't comprehensive: I asked and looked around, but didn't receive updates from everyone. I would be glad to learn of anything I've missed.)

When the new ToR are finalized (2023?) and the up-to-date membership list is made available, perhaps we can revisit the question of a structured post-detection working group (or groups) with clear objectives.

2021's Post-Detection update is [available here](#): much of what was said there is still applicable.

Some things haven't changed lately:

- the Committee has some unresolved issues concerning potential Protocol updates, *e.g.* best practices in relation to post-detection in today's information environment, given new and emerging search strategies and potential detection types, and the expanding community of searchers – all of which are vastly different now from when the Protocols were first formulated (shortly before the emergence of the WWW).
- addressing these issues comprehensively will require background research, including the integration of expertise which is currently not well-represented within the Committee itself.
- it bears repeating that post-detection related research is essential, but persistently underfunded.

And, of course, outbound transmissions from Earth – whether *de novo* or reply – continue to be a subject of debate, and at times friction, in this research community. Constructive ways forward should be sought.

Of particular note this year:

- Standards of evidence for biosignature detection / confidence in life detection is a burgeoning discussion in the astrobiology community, and this has intersections with technosignatures
- **mis/disinformation online is a serious and escalating concern** (discussed at some length in the PSETI Symposium Post-detection breakout session) which we should begin to address

Overview of today's update

1. Papers on themes related to Post-detection in 2022 IACs

2. Some relevant recent and upcoming activities of SETI Permanent Committee members including:

- SETI Institute and Breakthrough Listen, continued public outreach etc.
- UKSRN: Conference and Post-Detection Hub about to launch (St. Andrews)
- PSETI activities
- Confidence in Life Detection (CoLD) scale and Standards of Evidence for Biosignatures discussions
- Daniela de Paulis et al, A Sign in Space

3. Some Conferences and Symposia with post-detection relevance

- PSETI Symposium 2022 breakout session: "Post-detection for the 2020s: fundamentals, current concerns, and new discussions"
- COSPAR 2022
- AbSciCon 2022
- SSOCIA 2022
- Islamic Perspectives on Exotheology 2022

4. Relevant Research Centres and Seminar Series (established and emerging)

- The Technosignatures Seminar Series at NASA Goddard
- The Post-Detection Hub: <https://seti.wp.st-andrews.ac.uk/>
- Exploring Uncertainty and Risk in Contemporary Astrobiology (EURiCA), Durham University, UK
- The Centre for Cosmic Impact (SETI @ Durham)
- Würzburg University. Interdisciplinary Research Centre for Extraterrestrial Studies

5. A few noteworthy publications

6. A 'SETI in the Media' Case Study for 2022: FAST, June 2022

1. Post-detection Relevant Abstracts from IAC 2022, 19 Sept 2022, include ...

from <https://iafastro.directory/iac/browse/IAC-22/A4/>

Of course, the full range of papers in the SETI sessions, including the technical ones, is of relevance to post-detection, in terms of the forms of detection which are now or imminently possible. Some which more specifically focus on topics related to public reaction and related policy etc. include:

- Daniela De Paulis, A Sign in Space: an interdisciplinary exploration of the potential reception of an extraterrestrial signal. <https://iafastro.directory/iac/paper/id/67928/summary/> (Billingham Cutting-Edge Lecture)
 - Claudio Maccone, Fifty Years of SETI in the IAF Digital Library. <https://iafastro.directory/iac/paper/id/73236/summary/>
 - Ugur Guven, Spaceships in SETI Research and Potential for Spaceflight Technology. <https://iafastro.directory/iac/paper/id/67299/summary/>
 - Suresh Bhattari and Manisha Dwa, Are we alone? People's perception on an ExoWorld, its planet and possible life on it. <https://iafastro.directory/iac/paper/id/71922/summary/>
 - Mclee Kerolle, Ridha Aditya Nugraha, Sebastian Hernandez, Pragati Mot, The Legal Implications of Belonging to an Extraterrestrial Polity. <https://iafastro.directory/iac/paper/id/73495/summary/>
 - Léanne Fortuna, The World Configuration of Aliens Were Our Neighbours. <https://iafastro.directory/iac/paper/id/69479/summary/>
 - Carol Oliver, Astrobiology immersive Virtual Field Trip creation now as easy as PowerPoint. <https://iafastro.directory/iac/paper/id/71127/summary/>
 - Rebecca Charbonneau, CETI Across the Iron Curtain. <https://iafastro.directory/iac/paper/id/71778/summary/>
 - Gabriel G. De la Torre and Celia Avila-Rauch, Romanticism in Science as a form of cognitive bias and SETI. <https://iafastro.directory/iac/paper/id/73175/summary/>
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2. Some relevant recent and upcoming activities of SETI Permanent Committee members

The SETI Institute and Breakthrough Listen continue their extensive public outreach efforts.

The UK SETI Research Network (UKSRN), meeting and imminent Post-Detection Hub

(John Elliott and Martin Dominik will say more)

- **The Post-Detection Hub**, a UKSRN initiative, is on the verge of launching. See: <https://seti.wp.st-andrews.ac.uk/> At present, it is jointly hosted by the [St Andrews Centre for Exoplanet Science](#) and the [Centre for Global Law and Governance](#). Additional affiliates to be announced.
- The annual conference was this year organized/hosted by Michael Bohlander at Durham in July, and included papers relevant to post-detection by Michael Bohlander, Kate Genevieve, and John Elliott. www.durham.ac.uk/departments/academic/law/news-and-events/events/the-uk-seti-research-network--annual-conference-2022/

PSETI (Penn State) activities include (Jason Wright and Macy Huston will say more?)

PSETI 2022 symposium included “SETI & Society” topics and post-detection (more below).

SETI Bibliography project: See Lafond, Wright, and Huston: <https://arxiv.org/abs/2107.02887>

SETI-News. Macy Huston and Jason Wright provide this very valuable service: <https://seti.news/about/>

Macy Huston and Jason Wright, 2022. “[SETI in 2021](#)”. Helpful review includes papers on the social aspects of the search.

Confidence in Life Detection (CoLD) scale and Standards of Evidence for Biosignatures

Substantial discussion has occurred over the past 18 months (perhaps mostly in America / the NASA-sphere) regarding the CoLD scale and Standards of Evidence for Biosignatures, involving some members of the technosignatures community. It sounds as though there are future workshops in the planning stages, including more interdisciplinary researchers (journalism, social media, science communication, social science, etc.) This is all highly relevant to post-detection, announcement of life detection, etc. To be continued...

- Green, J. et al 2021. [Call for a framework for reporting evidence for life beyond Earth](#). Nature 598: 575–579.
- Meadows, V., H. Graham et al. 2022. [Community Report from the Biosignatures Standards of Evidence Workshop](#).
- [NExSS & NfoLD Standards of Evidence Joint Community Workshop description and talks](#).

- NASA Committee on Astrobiology and Planetary Science review of report, July 2022: [Independent Review of the Community Report from the Biosignature Standards of Evidence Workshop](#)
- N.b. so far, there has been minimal interdisciplinary input from specialists in the social sciences, history of science, science communication research, etc. If you are interested in the interdisciplinary side of this, see K. Denning's brief AbSciCon 2022 talk, "On Scales and Announcements of Life Detection: Relevant Input from Social Sciences Concerning Public Understandings of Science and Communication Strategies" ([access here](#)) and the resources from the PSETI Post-detection breakout session (see below).

Daniela de Paulis and team, A Sign in Space

This project, presented by PC member de Paulis during last year's PC meeting and again this year in the SETI sessions as the Billingham Cutting Edge Lecture, involves considerable discussion with an advisory group regarding post-detection issues and the public. De Paulis is also a [SETI Institute Artist in Residence](#).

3. Some Conferences and Symposia with post-detection relevance

The First Penn State SETI Symposium (June 27–30, 2022, State College, PA) involved some members of the PC (planning, speaking, and attending), and included a variety of talks, mostly technical, but also Rebecca Charbonneau on "Review: Social Sciences and SETI", Hilding Neilson on "The Drake Equation and the Search for Life through Indigenist Lenses", and Chelsea Haramia on "Is there a Difference Between Messaging and Mere Detectability?", as well as some posters with post-detection relevance. [Abstracts and talk recordings are available here, and posters too.](#)

The [Breakout sessions](#) included two interdisciplinary sessions, one specifically on Post-Detection, which some PC members attended (in person and remotely).

- **"Post-detection for the 2020s: fundamentals, current concerns, and new discussions"** was run as a hybrid session by Kathryn Denning, Brian McConnell, Rebecca Charbonneau, Chelsea Haramia, George Profitiliotis, with additional presenters Jill Tarter, Shelley Wright. Total: six hours of talks and active discussion exercises focused on Post-Detection themes, including astronomy, social science, history, communication systems engineering, methods from philosophy and Futures Studies.
 - [Full "Post-detection for the 2020s" breakout program with abstracts](#) .
 - The Post-detection Breakout session **talk recordings will be posted soon** (probably within a week or so) to the PSETI symposium website and the Zenodo page. Email kdenning@yorku.ca for more information and exact links.
 - **A substantial Google Drive folder of post-detection resources** was created for participants. It includes many papers from the PC's history, and some current resources. Please use it. PC members who would like access can [click here](#) and request it, or email kdenning@yorku.ca
- PSETI Dates for 2023: in the June 18-23 window. Hopefully there will be another post-detection session.

COSPAR in Athens in July

<https://www.cosparathens2022.org/> included technosignatures talks and launched the new COSPAR panel on Social Sciences and Humanities.

- [COSPAR panel on Social Sciences and Humanities](#)
- Session: Quest to Detect Extraterrestrial Life: Scientific Approaches and Cultural Impact of Discovery, organized by Nectaria Gizani and Giorgos P. Veldes. [Titles and abstracts](#). Talks of specific post-detection relevance included:
 - George D. Kyriakopoulos, In Search of Technosignatures and Biosignatures: challenges for international law.
 - Anuriti Saxena, Quest to Detect Extraterrestrial Life: Scientific Approaches and Cultural Impact of Discovery

AbSciCon, Atlanta, May 2022 (hybrid)

Relevant sessions included:

- [Scientists, Publics, Planetary Protection and Post-Detection: Urgent Discussions for the 2020s](#), with biosig/technosig-relevant talks from (follow links for abstracts) [Chelsea Haramia](#), [Clément Vidal](#), and [George Profitiliotis](#), and [Kathryn Denning](#).
- [Priorities and Strategies for Technosignature Detection](#), convened by Jacob Haqq-Misra, Sofia Sheikh, Ravi Kopparapu, and Clément Vidal.

SSOCIA (Society for Social and Conceptual Issues in Astrobiology), March 2022

- Included talks on SETI topics from: William Alba, Chelsea Haramia and Julia DeMarines, Daniela de Paulis, George Profitiliotis, Kathryn Denning, Tertia Gillet, Margarida Hermida, Walter Barta, Keith Abney, Dan Wueste, John Elliot, Carl DeVito, Lucas Mix, Seth Shostak.

Islamic Perspectives on Exotheology, online conference 10-11 May 2022

Convened by Shoaib Ahmed Malik and Jörg Matthias Determann, this covered a wide range of topics. [Conference booklet](#).

4. Research Centres and Seminar Series

The Technosignatures Seminar Series at NASA Goddard continues to feature speakers on a variety of technosignatures topics, including some from SETI PC members, and some directly relevant to post-detection e.g., Jim Green on 14 Sept 2022 on “The Confidence of Life Detection Scale”, and Jill Tarter on 21 Sept 2022 on “When Is The Evidence Sufficient To Claim The Discovery of Extraterrestrial Life?”. [Website with past and future speaker list and recordings of presentations.](#) To join the mailing list, contact Ravi: ravikumar.kopparapu@nasa.gov

A number of university centres have emerged recently / are emerging which address philosophical and/or social issues related to astrobiology and SETI. They include, for example:

- **The Post-Detection Hub (noted above):** <https://seti.wp.st-andrews.ac.uk/>
 - **Exploring Uncertainty and Risk in Contemporary Astrobiology (EURiCA)**, Durham University, UK, at the Centre for Humanities Engaging Science and Society (CHESS): www.durham.ac.uk/research/institutes-and-centres/humanities-engaging-science-society/research/eurica-project-leverhulme/, including a seminar series on Philosophical Issues in Space Science: www.durham.ac.uk/research/institutes-and-centres/humanities-engaging-science-society/research/eurica-project-leverhulme/philosophical-issues-in-space-science-seminar-series/
 - Also at Durham: **The Centre for Cosmic Impact (SETI @ Durham)**: Not much information available yet, but Dr Hermine Landt-Wilman presented on it at BIS in July. John Elliott and Michael Bohlander are also connected. www.bis-space.com/event/space-law-and-alien-life-combined-approaches-to-astrobiology-law-and-the-search-for-extra-terrestrial-intelligence/
 - **Würzburg University. Interdisciplinary Research Centre for Extraterrestrial Studies**, led by Hakan Kayal. www.uni-wuerzburg.de/en/ifex/home/
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5. A few noteworthy publications

(there are more out there...)

Jensine Andresen and Octavio Chon Torres, eds. *Extraterrestrial Intelligence: Academic and Societal Implications*. Cambridge Scholars, 2022. www.cambridgescholars.com/product/978-1-5275-7727-5

Anamaria Berea, ed. 2022. [Technosignatures for Detecting Intelligent Life in Our Universe: A Research Companion - Astrobiology Perspectives on Life in the Universe](#). Wiley. Includes chapters on post-detection topics.

Professor Michael Bohlander of Durham University Law School has been highly active in this field over the past couple of years. In April 2022 he became their Chair in Global Law and SETI Policy. An overview of his work is [here](#). Upcoming book: *Contact with Extraterrestrial Intelligence and Human Law* (Springer).

Ian Crawford, ed. 2021. [Expanding Worldviews: Astrobiology, Big History and Cosmic Perspectives](#). Springer.

Octavio Chon Torres, Ted Peters, Joseph Seckbach, Richard Gordon, eds. 2021. [Astrobiology: Science, Ethics, and Public Policy](#). Wiley.

Jörg Matthias Determann. [Islam, Science Fiction and Extraterrestrial Life: The Culture of Astrobiology in the Muslim World](#). London: I.B. Tauris, 2020.

Steven J Dick, 2020. [Space, Time, and Aliens: Collected Works on Cosmos and Culture](#) (Springer).

Chelsea Haramia, forthcoming 2023: "Understanding the Question of Whether to Message Extraterrestrial Intelligence", in *Routledge Handbook of Social Studies of Outer Space*.

Macy Huston and Jason Wright, 2022. ["SETI in 2021"](#). Helpful review includes papers on the social aspects of the search.

Brian McConnell. 2021. [The Alien Communication Handbook: So We Received a Signal—Now What?](#) Springer. See also McConnell's essays at Centauri Dreams: [first](#), [second](#), and [third](#).

George Profitiliotis. 2022. ["Policy Memo: Anticipatory Science Diplomacy as a Governance and Cooperation Mechanism for the Search for Extraterrestrial Life."](#) Journal of Science Policy & Governance

[Settler Science, Alien Contact, and Searches for Intelligence](#), American Indian Culture and Research Journal Vol 45 No 1, Dec 2021, edited by David Delgado Shorter and Kim TallBear, includes multiple papers from Indigenous studies scholars and other interdisciplinary researchers reflecting on SETI and contact. The critical perspective through an Indigenous lens has been sparking conversation.

Jason Wright, Chelsea Haramia, and Gabriel Swiney. Forthcoming. "Geopolitical Implications of a Successful SETI Program"). Space Policy. This paper is a response to Wisian & Traphagan (2020) ["The Search for Extraterrestrial Intelligence: A Realpolitik Consideration"](#), Space Policy 52.

6. A 'SETI in the Media' Case Study for 2022: FAST: premature/confused press / social media flurry in June 2022

(Some observations from surveying English language news sites, compiled June 2022. Not comprehensive... It would be interesting to see differences internationally / in different languages.)

As is generally the case with potential signals, this progressed very quickly to widespread media coverage **within a day**. The range of headlines and content was significant: even in first-generation articles (let alone those derived from them), there was a wide range of descriptions of what had happened, and of the significance and implications. And the accompanying illustrations varied from the classic grey alien to the FAST dish, etc.

Each incident like this provides valuable data about how information (and misinformation) would flow in the event of a strong candidate or confirmed detection. (But we need proper studies of this!)

- the English (Google) translation of the original June 14 report in *Science and Technology Daily* (the Chinese media outlet) which broke the story: [SciTech Daily 14 June 2022 Eng translation](#) "Searching for extraterrestrial civilization, "China Sky Eye" found suspicious signals".
- 14 June Bloomberg article that seems to have kicked off the fuss in the English speaking press etc [China Says Giant Telescope May Have Detected Signals From Alien Civilizations - Bloomberg](#)
- Rapid response from some SETI scientists including Dan Werthimer who was quoted in multiple interviews within the first 2 days..
- [Did a giant radio telescope in China just discover aliens? Not so FAST...](#) June 16, The Conversation, by Danny Price of BL. Good summary and also an example of an effective rapid media response that the BL team does well.
- Dan Werthimer's interview with *Futurism* (June 16) [Sorry, But New Signal Is Definitely Not Aliens, Says Scientist Working on Project](#) ... this article represents the situation as a difference of opinion between scientists..... and also linked to the then-not-yet-peer-reviewed preprint [The most sensitive SETI observation by multi-beam coincidence matching strategy towards exoplanet systems](#) ... Also interesting though: this new article was juxtaposed on the Futurism site with previous alien-related titles like "Study: There May Be as Many as Four Evil Civilizations in Our Galaxy", and "Head of NASA Suggests That UFOs Could Be Alien Technology" ... which would also presumably happen if this article was about a real detection, and that might matter for public reception.
- [No, a Chinese telescope didn't just detect aliens. Here's what actually happened | Salon.com](#) Nicole Karlis (June 15, includes interview with Dan Werthimer, comments from Avi Loeb and Seth Shostak) . This helpfully includes a link to the English (Google) translation of the original June 14 report in *Science and Technology Daily* (the Chinese media outlet)
- [A Chinese Telescope Did Not Find an Alien Signal. The Search Continues. - The New York Times](#) (pirated copy [A Chinese Telescope Did Not Find an Alien Signal. The Search Continues. - The New York Times | Canada News Media](#)) "China's astronomers have been initiated into the search for extraterrestrial intelligence with the kind of false alarm that others in the field have experienced for decades...This has become a familiar story. For half a century, SETI, or the search for extraterrestrial intelligence, has been a game of whack-a-mole, finding promising signals before tracking them down to orbiting satellites, microwave ovens and other earthly

sources. Dr. Drake himself pointed a radio telescope at a pair of stars in 1960 and soon thought he had struck gold, only to find out the signal was a stray radar..”

- Inverse, Rahul Rao, 16 June: [China's search for alien life turns up something suspicious — but there's a catch](#) - quotes Seth Shostak.
- Some description of the Twitter flurry [EarthSky | China's FAST telescope reports signal from aliens, then deletes report](#) , Byrd and Anderson
- [If aliens are calling, let it go to voicemail](#) , Bryan Walsh, Vox, 18 June. integrates this latest story with points about existential risk, starting with [A for Andromeda](#) (Fred Hoyle story).
- [No, China's 'Sky Eye' Telescope Has Not Detected Radio Signals From Alien Civilizations - CNET](#) , Jackson Ryan comments on the risks of premature/not real announcements/leaks like this.

For some context: Dr Di Li gave a great update here about FAST activities including SETI, Feb 2021: www.youtube.com/watch?v=GOcJ14HA4Xg in the IAF session on “Breakthrough Listen and International SETI Collaborations”.
